

Anindya Mondal

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PhD Candidate (AI) — Research focus: VLMs, controllable generation, agentic feedback loops.

EDUCATION

University of Surrey

Guildford, UK

PhD Candidate in Artificial Intelligence

Oct 2022 – Present

- Surrey Institute for People-Centred AI, CVSSP. Research on vision–language integration, controllable generative models, and VLM-guided agentic workflows.

Jadavpur University

Kolkata, India

B.E. (Hons.) in Electronics & Telecommunication Engineering

Aug 2018 – Jun 2022

- GPA: 8.79/10

SELECTED PROJECTS

ABACUS (Under Review)

2025 – 2026

Unified Foundation Model for Image Counting

University of Surrey

- Using post-training to internalise high-instance count understanding directly into a unified VLM, enabling count-aware generation and visual reasoning without external verification loops.

CountLoop (Preprint)

2024 – 2025

VLM agent-guided high-instance image generation

University of Surrey

- Built a VLM agent that iteratively closes the loop with a diffusion generator to control instance counts and spatial layout in high-instance scenes — without retraining the generator.

Action-Count Faithful Video Generation

2025 – 2026

VLM-guided trajectory control for generative video models

University of Surrey

- Extends the agentic feedback paradigm to the temporal domain: uses a VLM to guide trajectory planning inside a video generation model so that actions occur the correct number of times across frames.

OmniCount (AAAI 2025)

2024 – 2025

Class-agnostic multi-label object counting

University of Surrey

- Counting model leveraging semantic and geometric priors for robust generalisation across crowded, high-instance scene categories with vision–language feedback integration.

MSQNet (ICCVW 2023)

2023

Multimodal Transformer for Action Recognition

University of Surrey

- Transformer-based multimodal action-recognition model using multi-modal query fusion; significant precision gains in actor-agnostic, class-agnostic action understanding.

RESEARCH EXPERIENCE

University of Surrey

Guildford, UK

Doctoral Researcher

Oct 2022 – Present

- Led research on VLM-guided agentic orchestration for controllability in generative models; developed automated evaluation loops for count verification and layout matching.

Indian Institute of Science

Bengaluru, India

Research Intern

May 2022 – Aug 2022

- Devised a source-free domain-adaptation framework for image classification under domain shift.

SELECTED PUBLICATIONS

UniCount: Unified Foundation Model for Image Counting.

A. Mondal et al. Under Preparation.

CountLoop: Iterative Agent Guided High Instance Image Generation.

A. Mondal, A. Banerjee, S. Nag, J. Lladós, X. Zhu, A. Dutta. Preprint. [Project]

OmniCount: Multi-label Object Counting with Semantic–Geometric Priors.

A. Mondal, S. Nag, X. Zhu, A. Dutta. AAAI 2025. DOI: 10.1609/aaai.v39i18.34151

Actor-Agnostic Multi-label Action Recognition with Multi-modal Query.

A. Mondal, S. Nag, J.M. Prada, X. Zhu, A. Dutta. ICCVW 2023. DOI: 10.1109/ICCVW60793.2023.00086

TECHNICAL SKILLS

- **Languages:** Python (expert), MATLAB (expert), C (advanced).
- **Frameworks:** PyTorch (expert), TensorFlow (familiar), scikit-learn, NumPy, SciPy, Pandas.
- **Agent/LLM tooling:** Tool-augmented VLM/LLM workflows, prompt engineering, evaluation loops, and programmatic tool invocation for verification and planning.
- **Other tools:** Git, Docker, Jupyter Notebook, L^AT_EX.

AWARDS AND FUNDING

AAAI 2025 Conference & Travel Grant, Philadelphia, USA.

ICCV 2023 Conference Grant, Paris, France.

Postgraduate Studentship, University of Surrey (2022).

Uplink Research Internship Award, ACM SIGKDD India Chapter (2022).

PROFESSIONAL EXPERIENCE & SERVICE

University of Surrey

Guildford, UK

Teaching Assistant / Teaching Fellow

2023 – 2025

- Demonstrator: Applied Machine Learning (EEEM068); Advanced Topics in Computer Vision & Deep Learning (EEEM071).
- Teaching Fellow: Applied Machine Learning, University of Surrey MSc People-Centred AI (edX).

Reviewer

Conferences & Journals

2022 – 2025

- CVPR, ICCV, ECCV, NeurIPS, ICASSP, BMVC, IEEE TSP, TSIPN, ISBI.